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INFORMATION PROCESSING SYSTEM, INFORMATION PROCESSING METHOD, AND NON-TRANSITORY COMPUTER READABLE STORAGE MEDIUM STORING INFORMATION PROCESSING PROGRAM

Abstract

An information processing system acquires moving body information that indicates a position of each of multiple moving bodies existing in the real world. The information processing system acquires, based on the moving body information, predictive moving body information at a prescribed cycle. The predictive moving body information indicates a predictive position of each of the multiple moving bodies after a point in time at which the moving body information is acquired. The information processing system is configured to set the cycle for acquiring the predictive moving body information in at least one zone among multiple zones in which the moving bodies are present to be longer than the cycle for acquiring the predictive moving body information in the other zones.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2024-030138, filed on Feb. 29, 2024, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Field

[0002] The present disclosure relates to an information processing system, an information processing method, and a non-transitory computer readable storage medium storing an information processing program for reproducing a real-world traffic environment in a virtual space.

2. Description of Related Art

[0003] Digital twin technology enables the reproduction of real-world environments within a virtual space. Japanese Laid-Open Patent Publication No. 2020-013557 discloses a system that utilizes a traffic digital twin to reproduce a real-world traffic environment in a virtual space.

[0004] To accurately replicate the real-world positions of multiple moving bodies in a virtual space, one approach is to minimize the update cycle for the positions of the moving bodies in the virtual world. However, this approach significantly increases the processing load on the device responsible for updating the positions of multiple moving bodies.

SUMMARY

[0005] This Summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used as an aid in determining the scope of the claimed subject matter.

[0006] In a first general aspect, an information processing system includes processing circuitry. The processing circuitry is configured to acquire moving body information that indicates a position of each of multiple moving bodies existing in the real world, and acquire, based on the moving body information, predictive moving body information at a prescribed cycle. The predictive moving body information indicates a predictive position of each of the multiple moving bodies after a point in time at which the moving body information is acquired. The processing circuitry is also configured to set the cycle for acquiring the predictive moving body information in at least one zone among multiple zones in which the moving bodies are present to be longer than the cycle for acquiring the predictive moving body information in the other zones.

[0007] In a second general aspect, an information processing method includes: acquiring moving body information that indicates a position of each of multiple moving bodies existing in the real world; acquiring, based on the moving body information, predictive moving body information at a prescribed cycle, the predictive moving body information indicating a predictive position of each of the multiple moving bodies after a point in time at which the moving body information is acquired; and setting the cycle for acquiring the predictive moving body information in at least one zone among multiple zones in which the moving bodies are present to be longer than the cycle for

acquiring the predictive moving body information in the other zones.

[0008] In a third general aspect, a non-transitory computer readable storage medium stores an information processing program to be executed by processing circuitry included in an information processing system. The information processing program causes the processing circuitry to set a cycle for acquiring predictive moving body information in at least one zone among multiple zones in which the moving bodies are present to be longer than the cycle for acquiring the predictive moving body information in the other zones. The predictive moving body information is acquired based on moving body information that indicates a position of each of multiple moving bodies present in the real world. The predictive moving body information indicates a predictive position of each of the multiple moving bodies after a point in time at which the moving body information is acquired.

[0009] Other features and aspects will be apparent from the following detailed description, the drawings, and the claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a schematic diagram illustrating an information processing system according to an embodiment.

[0011] FIG. 2 is a schematic diagram illustrating a processing device, a communication device, and a storage device included in the information processing system shown in FIG. 1.

[0012] FIG. 3 is a schematic diagram showing a manner in which the information processing system shown in FIG. 1 acquires predictive moving body information.

[0013] FIG. 4 is a flowchart showing a flow of processes executed by the information processing system shown in FIG. 1.

[0014] FIG. 5 is a schematic diagram illustrating an example of an update field for which the information processing system shown in FIG. 1 updates predictive moving body information.

[0015] FIG. 6 is a table showing a relationship between moving speeds of the moving bodies shown in FIG. 5, update cycle labels, and update cycles of predictive moving body information.

[0016] FIG. 7 is a table showing moving speeds, zones, and update cycles of the respective moving bodies in the update field shown in FIG. 5.

[0017] FIG. 8 is a table showing a manner in which the processing device included in the information processing system shown in FIG. 1 adjusts update cycles.

[0018] Throughout the drawings and the detailed description, the same reference numerals refer to the same elements. The drawings may not be to scale, and the relative size, proportions, and depiction of elements in the drawings may be exaggerated for clarity, illustration, and convenience.

DETAILED DESCRIPTION

[0019] This description provides a comprehensive understanding of the methods, apparatuses, and/or systems described. Modifications and equivalents of the methods, apparatuses, and/or systems described are apparent to one of ordinary skill in the art. Sequences of operations are exemplary, and may be changed as apparent to one of ordinary skill in the art, with the exception of operations necessarily occurring in a certain order. Descriptions of functions and constructions that are well known to one of ordinary skill in the art may be omitted.

[0020] Exemplary embodiments may have different forms, and are not limited to the examples described. However, the examples described are thorough and complete, and convey the full scope of the disclosure to one of ordinary skill in the art.

[0021] In this specification, “at least one of A and B” should be understood to mean “only A, only B, or both A and B.”

[0022] An information processing system **10** according to an embodiment will now be described

with reference to FIGS. 1 to 8.

Overview of Information Processing System 10

[0023] The information processing system **10** acquires moving body information, which is information indicating the position and behavior of each of multiple moving bodies **800** in the real world, at multiple points in time. The moving bodies **800** include vehicles **600**, pedestrians **700**, bicycles, and animals. Vehicles **600** include two-wheeled vehicles.

[0024] The information processing system **10** calculates, using the acquired moving body information, predictive moving body information indicating the predictive position of each moving body **800** after the point in time at which the moving body information is acquired, at a prescribed update cycle. The predictive moving body information is obtained by reproducing a real-world traffic environment in a virtual space.

[0025] As shown in FIG. 1, the information processing system **10** is configured to communicate, via an external communication network **400**, with multiple information processing terminals **500** (only one is shown), multiple vehicles **600** (only one is shown), and multiple roadside sensors **900**.

[0026] The information processing terminal **500** shown in FIG. 1 collects the location information of a pedestrian **700** carrying an information processing terminal **500** as moving body information. The location information is coordinate values of latitude and longitude. The information processing terminal **500** transmits the collected location information of the pedestrian **700** to the information processing system **10** via the external communication network **400**. The information processing terminal **500** is, for example, a smartphone or the like carried by the pedestrian **700**. The information processing terminal **500** may be a wearable terminal, a tablet terminal, or the like. Examples of the wearable terminal include a bracelet-type terminal worn on the wrist and a necklace-type terminal worn on the neck.

[0027] The vehicle **600** shown in FIG. 1 is equipped with vehicle on-board sensors **610**. The vehicle **600** transmits the moving body information collected by the vehicle on-board sensors **610** to the information processing system **10** via the external communication network **400**. The vehicle on-board sensors **610** include, for example, a vehicle speed sensor, an accelerator sensor, a brake sensor, a steering sensor, and an acceleration sensor. The acceleration sensor is, for example, an inertial measurement unit (IMU).

[0028] The vehicle **600** includes an external camera, a sonar sensor, and a location information acquisition system as the vehicle on-board sensors **610**. The external camera and the sonar sensor mounted on the vehicle **600** collect information on distances between the vehicle **600** and other objects located around the vehicle **600** to generate observation data. The vehicle **600** may include a light detection and ranging (LiDAR) sensor as a sensor that serves a role similar to that of an external camera and a sonar sensor. The location information acquisition system is, for example, a global navigation satellite system (GNSS), a real time kinematic (RTK) system, a LiDAR sensor, or the like.

[0029] The vehicle on-board sensors **610** collect moving body information, including vehicle information such as a vehicle identification number (VIN) of the vehicle **600** and information on a vehicle speed, a traveling direction, and a traveling trajectory and a position of the vehicle **600**, for example.

[0030] The roadside sensors **900** shown in FIG. 1 are multiple sensors installed on the road. The roadside sensors **900** include multiple traffic lights **910**, multiple roadside cameras **920**, and LiDAR sensors. Each traffic light **910** transmits information related to changes in the state of the traffic infrastructure, such as the timing at which the traffic light **910** turns green and the time during which the traffic light **910** remains green, to the information processing system **10** via the external communication network **400**.

[0031] Each roadside camera **920** collects observation data around the roadside camera **920**. The observation data includes moving body information of multiple moving bodies **800** around each roadside camera **920**. The roadside cameras **920** include, for example, visible light cameras and

infrared cameras. The LiDAR sensors installed on the road acquire point cloud data, arranged in chronological order, that is continuously observed at fixed time intervals. Each LiDAR sensor collects the moving body information of moving bodies **800** located around the LiDAR sensor.

Configuration of Information Processing System **10**

[0032] As illustrated in FIG. **2**, the information processing system **10** includes a processing device **100**, a storage device **200**, and a communication device **300**.

[0033] The processing device **100** includes first processing circuitry **101**, first storage circuitry **102**, and first communication circuitry **103**. Programs are stored in the first storage circuitry **102**. The first processing circuitry **101** executes programs stored in the first storage circuitry **102** to execute various kinds of processes. The first processing circuitry **101** includes a processor. The processing device **100** is connected to the external communication network **400** via the first communication circuitry **103**.

[0034] The storage device **200** includes second processing circuitry **201**, second storage circuitry **202**, and second communication circuitry **203**. Programs are stored in the second storage circuitry **202**. The second processing circuitry **201** executes programs stored in the second storage circuitry **202** to execute various kinds of processes. The second processing circuitry **201** includes a processor. The storage device **200** is connected to the external communication network **400** via the second communication circuitry **203**.

[0035] The communication device **300** includes third processing circuitry **301**, third storage circuitry **302**, and third communication circuitry **303**. Programs are stored in the third storage circuitry **302**. The third processing circuitry **301** executes programs stored in the third storage circuitry **302** to execute various kinds of processes. The third processing circuitry **301** includes a processor. The communication device **300** is connected to the external communication network **400** via the third communication circuitry **303**.

[0036] Each of the processing device **100**, the storage device **200**, and the communication device **300** includes processing circuitry including one or more processors that execute various processes in accordance with a computer program (software). Each of the processing device **100**, the storage device **200**, and the communication device **300** may include processing circuitry including one or more dedicated hardware circuits such as an application-specific integrated circuit (ASIC) that executes at least a part of various processes. Alternatively, each of the processing device **100**, the storage device **200**, and the communication device **300** may include processing circuitry including a combination of one or more processors and one or more dedicated hardware circuits. Each processor includes a CPU and a memory module, such as a RAM and a ROM, and the memory module stores program codes or instructions configured to cause the CPU to execute processes. The memory module, which is a non-transitory computer-readable storage medium, includes any type of media that are accessible by general-purpose computers and dedicated computers.

[0037] The configuration of the information processing system **10** is not limited to the configuration illustrated in FIG. **2**. For example, the processing device **100**, the storage device **200**, and the communication device **300** may be included in a single server. For example, the processing device **100**, the storage device **200**, and the communication device **300** may be connected to each other via wired connections in a manner that allows mutual communication.

[0038] FIG. **2** illustrates a first vehicle **601** and a second vehicle **602** as examples of the moving bodies **800** for which the information processing system **10** calculates the predictive moving body information. The first vehicle **601** includes a first vehicle on-board sensor **611**. The first vehicle on-board sensor **611** transmits the moving body information of the first vehicle **601** to the information processing system **10** via the external communication network **400**. The second vehicle **602** includes a second vehicle on-board sensor **612**. The second vehicle on-board sensor transmits the moving body information of the second vehicle **602** to the information processing system **10** via the external communication network **400**.

[0039] The communication device **300** acquires the moving body information transmitted from

sensors at a prescribed acquisition cycle via the third communication circuitry **303**. The communication device **300** stores the acquired moving body information in the third storage circuitry **302**. The third processing circuitry **301** of the communication device **300** transmits the moving body information stored in the third storage circuitry **302** to the processing device **100** via the third communication circuitry **303**.

[0040] The processing device **100** acquires the moving body information from the communication device **300** via the first communication circuitry **103**. The processing device **100** stores the received moving body information in the first storage circuitry **102**. The first processing circuitry **101** of the processing device **100** calculates predictive moving body information using the moving body information. The processing device **100** transmits the predictive moving body information to the storage device **200** via the first communication circuitry **103**.

[0041] The storage device **200** receives the predictive moving body information via the second communication circuitry **203**. The storage device **200** stores the acquired predictive moving body information in the second storage circuitry **202**. The second processing circuitry **201** of the storage device **200** provides the predictive moving body information stored in the second storage circuitry **202** in response to a request via the external communication network **400**.

[0042] The first vehicle **601** and the second vehicle **602** are examples of the moving bodies **800** for which the information processing system **10** calculates predictive moving body information. For example, the information processing system **10** calculates the predictive moving body information of the first vehicle **601** based on the moving body information acquired from the first vehicle on-board sensor **611**. For example, the information processing system **10** calculates the predictive moving body information of the second vehicle **602** based on the moving body information acquired from the second vehicle on-board sensor **612**.

Calculation of Predictive Moving Body Information by Information Processing System **10**

[0043] A manner in which the information processing system **10** calculates predictive moving body information will now be described with reference to FIG. **3**. The communication device **300** of the information processing system **10** acquires the moving body information of each moving body **800** at the prescribed acquisition cycle. For example, the communication device **300** of the information processing system **10** acquires the moving body information of the first vehicle **601** from the first vehicle on-board sensor **611** of the first vehicle **601** every 0.1 seconds. In other words, the communication device **300** acquires the moving body information of the first vehicle **601** from the first vehicle on-board sensor **611** at an acquisition cycle of 0.1 seconds.

[0044] In FIG. **3**, the moving body information of the first vehicle **601** acquired by the communication device **300** at the acquisition cycle of 0.1 seconds is represented as first moving body information DAT_1, second moving body information DAT_2, third moving body information DAT_3, fourth moving body information DAT_4, fifth moving body information DAT_5, and sixth moving body information DAT_6. The first moving body information DAT_1 is the moving body information of the first vehicle **601** acquired at a point in time T0. The second moving body information DAT_2 is the moving body information of the first vehicle **601** acquired at a point in time T1, which is 0.1 seconds after the point in time T0. The third moving body information DAT_3 is the moving body information of the first vehicle **601** acquired at a point in time T2, which is 0.1 seconds after the point in time T1. The fourth moving body information DAT_4 is the moving body information of the first vehicle **601** acquired at a point in time T3, which is 0.1 seconds after the point in time T2. The fifth moving body information DAT_5 is the moving body information of the first vehicle **601** acquired at a point in time T4, which is 0.1 seconds after the point in time T3. The sixth moving body information DAT_6 is the moving body information of the first vehicle **601** acquired at a point in time T5, which is 0.1 seconds after the point in time T4.

[0045] The acquisition cycle does not need to be a common cycle across the multiple moving bodies **800**. For example, the communication device **300** acquires the moving body information of

the second vehicle **602** from the second vehicle on-board sensor **612** of the second vehicle **602** every 0.5 seconds. In other words, the communication device **300** acquires the moving body information of the second vehicle **602** from the second vehicle on-board sensor **612** at an acquisition cycle of 0.5 seconds.

[0046] In FIG. **3**, the moving body information of the second vehicle **602** acquired by the communication device **300** at the acquisition cycle of 0.5 seconds is shown as seventh moving body information DAT_7 and eighth moving body information DAT_8. The seventh moving body information DAT_7 is moving body information of the second vehicle **602** acquired at the point in time T0. The eighth moving body information DAT_8 is moving body information of the second vehicle **602** acquired at the point in time T5, which is 0.5 seconds after the point in time T0.

[0047] The third storage circuitry **302** of the communication device **300** shown in FIG. **2** stores the acquired moving body information. The communication device **300** transmits the moving body information stored in the third storage circuitry **302** to the processing device **100** via the third communication circuitry **303**.

[0048] When acquiring the moving body information from the communication device **300** via the first communication circuitry **103**, the processing device **100** updates the predictive moving body information of the first vehicle **601** and the second vehicle **602** every 0.1 seconds based on the acquired moving body information. First predictive moving body information PRE_1, second predictive moving body information PRE_2, third predictive moving body information PRE_3, fourth predictive moving body information PRE_4, fifth predictive moving body information PRE_5, and sixth predictive moving body information PRE_6 shown in FIG. **3** are predictive moving body information of the first vehicle **601** and the second vehicle **602** updated every 0.1 seconds.

[0049] The processing device **100** transmits the updated predictive moving body information to the storage device **200** via the first communication circuitry **103**. When acquiring the predictive moving body information via the second communication circuitry **203**, the storage device **200** stores the acquired predictive moving body information in the second storage circuitry **202**.

[0050] When the information processing system **10** does not determine an appropriate update cycle of the predictive moving body information for each of the moving bodies **800**, the update cycle of the predictive moving body information and the acquisition cycle of the moving body information can be different from each other. When the update cycle of the predictive moving body information is different from the acquisition cycle of the moving body information, the processing device **100** needs to calculate calculation data CAL, which is the location information of the moving body **800** at a time when no moving body information of the moving body **800** is available, based on the moving body information of the moving body **800**.

[0051] As shown in FIG. **3**, when the communication device **300** acquires the moving body information of the second vehicle **602** at the point in time T0, the next point in time at which the communication device **300** acquires the moving body information of the second vehicle **602** is the point in time T5, which is 0.5 seconds later. Therefore, the processing device **100** calculates the calculation data CAL, which represents the location information of the second vehicle **602** at a time when no moving body information of the second vehicle **602** is available, based on the moving body information of the second vehicle **602**.

[0052] As shown in FIG. **3**, the processing device **100** calculates first calculation data CAL_1, which is the location information of the second vehicle **602** at the point in time T0, based on the seventh moving body information DAT_7 of the second vehicle **602** at the point in time T1 stored in the first storage circuitry **102**. The processing device **100** calculates second calculation data CAL_2, which is the location information of the second vehicle **602** at the point in time T2, based on the seventh moving body information DAT_7. The processing device **100** calculates third calculation data CAL_3, which is the location information of the second vehicle **602** at the point in time T3, based on the seventh moving body information DAT_7. The processing device **100**

calculates fourth calculation data CAL_4, which is the location information of the second vehicle **602** at the point in time T4, based on the seventh moving body information DAT_7. The first processing circuitry **101** of the processing device **100** calculates the predictive moving body information based on the moving body information and the calculation data CAL.

[0053] As shown in FIG. 3, enclosed by a long-dash double-short-dash line, the processing device **100** calculates the first predictive moving body information PRE_1 based on the first moving body information DAT_1 and the seventh moving body information DAT_7 at the point in time T0. At the point in time T1, the processing device **100** calculates the second predictive moving body information PRE_2 based on the second moving body information DAT_2 and the first calculation data CAL_1. At the point in time T2, the processing device **100** calculates the third predictive moving body information PRE_3 based on the third moving body information DAT_3 and the second calculation data CAL_2. At the point in time T3, the processing device **100** calculates the fourth predictive moving body information PRE_4 based on the fourth moving body information DAT_4 and the third calculation data CAL_3. At the point in time T4, the processing device **100** calculates the fifth predictive moving body information PRE_5 based on the fifth moving body information DAT_5 and the fourth calculation data CAL_4. At the point in time T5, the processing device **100** calculates the sixth predictive moving body information PRE_6 based on the sixth moving body information DAT_6 and the eighth moving body information DAT_8.

[0054] When the update cycle at which the information processing system **10** updates the predictive moving body information is relatively short, the amount of the predictive moving body information to be updated by the information processing system **10** per unit time increases. That is, when the prescribed update cycle, at which the information processing system **10** updates the predictive moving body information, is relatively short, the processing load on the information processing system **10** increases. In this regard, the information processing system **10** determines an appropriate update cycle of the predictive moving body information for each moving body **800** by executing the following processes.

Processes Executed by Information Processing System **10**

[0055] Hereinafter, with reference to FIGS. 4 to 8, a flow of processes in which the information processing system **10** determines the update cycle at which the predictive moving body information is updated will be described.

[0056] FIG. 4 is a flowchart showing a flow of a series of processes executed by the information processing system **10**. The information processing system **10** repeatedly executes this series of processes.

Acquisition of Moving Body Information

[0057] As shown in FIG. 4, when the series of processes is started, the information processing system **10** first acquires the moving body information in an update field **20** in step S10.

Specifically, the communication device **300** acquires the moving body information in the update field **20** shown in FIG. 5 via the third communication circuitry **303**. The moving body information acquired by the information processing system **10** includes location information of each moving body **800**, a moving speed V of each moving body **800**, a traveling direction of each moving body **800**, and a type of each moving body **800**.

[0058] FIG. 5 illustrates nine moving bodies **800**, which are referred to respectively as a first moving body **801**, a second moving body **802**, a third moving body **803**, a fourth moving body **804**, a fifth moving body **805**, a sixth moving body **806**, a seventh moving body **807**, an eighth moving body **808**, and a ninth moving body **809**. The first moving body **801**, the third moving body **803**, the fifth moving body **805**, and the eighth moving body **808** are vehicles **600**. The second moving body **802**, the fourth moving body **804**, the sixth moving body **806**, the seventh moving body **807**, and the ninth moving body **809** are pedestrians **700**. The information processing system **10** acquires the moving body information of multiple moving bodies **800** in the update field **20** from multiple sensors. After acquiring the moving body information in the update field **20**, the communication

device **300** advances the process to step **S11**.

Division of Update Field **20**

[0059] In step **S11**, the processing device **100** of the information processing system **10** acquires the moving body information of each moving body **800** from the communication device **300**. In step **S11**, the processing device **100** establishes a zone for each moving body **800** based on the location information, the moving speed V , and the traveling direction of each moving body **800**.

Specifically, the processing device **100** establishes multiple zones within the update field **20**, corresponding to each moving body **800**, such that only one moving body **800** is present within each zone. The range of the zone established for each moving body **800** encompasses the predicted range within which the moving body **800** is expected to be located.

[0060] As shown in FIG. 5, the processing device **100** establishes a first zone **AR_1** for the first moving body **801**. The processing device **100** establishes a second zone **AR_2** for the second moving body **802**. The processing device **100** establishes a third zone **AR_3** for the third moving body **803**. The processing device **100** establishes a fourth zone **AR_4** for the fourth moving body **804**. The processing device **100** establishes a fifth zone **AR_5** for the fifth moving body **805**. The processing device **100** establishes a sixth zone **AR_6** for the sixth moving body **806**. The processing device **100** establishes a seventh zone **AR_7** for the seventh moving body **807**. The processing device **100** establishes an eighth zone **AR_8** for the eighth moving body **808**. The processing device **100** establishes a ninth zone **AR_9** with respect to the ninth moving body **809**.

[0061] The processing device **100** determines the size of the zone to be established for each moving body **800** according to the moving speed V of the moving body **800** acquired from the communication device **300**. Specifically, the processing device **100** narrows the zone assigned to each moving body **800** as the moving speed V of the moving body **800** decreases.

[0062] As shown in FIG. 7, the moving speed V of the first moving body **801** is 50 kilometers per hour. The moving speed V of the eighth moving body **808** is 20 kilometers per hour. Accordingly, as shown in FIG. 5, the processing device **100** sets the eighth zone **AR_8** established for the eighth moving body **808** to be narrower than the first zone **AR_1** established for the first moving body **801**.

[0063] The processing device **100** may change the range of the zone established for each moving body **800** according to the type of the moving body **800**. As illustrated in FIG. 5, the processing device **100** sets zones established for the second moving body **802**, the fourth moving body **804**, the sixth moving body **806**, the seventh moving body **807**, and the ninth moving body **809**, which are pedestrians **700**, to be narrower than zones established for the first moving body **801**, the third moving body **803**, the fifth moving body **805**, and the eighth moving body **808**, which are vehicles **600**.

[0064] The processing device **100** calculates a predicted path based on the acquired information on the traveling direction of each moving body **800**. The processing device **100** establishes a zone such that the zone is wider on the side on which the predicted path of the moving body **800** extends. For example, the predicted path of the fifth moving body **805** extends forward and rightward from the fifth moving body **805**. The processing device **100** establishes the fifth zone **AR_5** for the fifth moving body **805** such that the fifth zone **AR_5** is wider on the side on which the predicted path extends. The centroid of the fifth zone **AR_5** established for the fifth moving body **805** is located on the side on which the predicted path of the fifth moving body **805** extends with respect to the center of the fifth moving body **805**.

[0065] The processing device **100** may omit calculating the predictive moving body information for the moving body **800** in areas of the real world where the moving bodies **800** are assumed to be unable to enter. The processing device **100** may omit establishing zones for the moving bodies **800** in ranges where the moving bodies **800** are assumed to be unable to enter. As shown in FIG. 5, the processing device **100** establishes the zone for each moving body **800** such that the zone does not overlap with the shaded regions, which correspond to zones occupied by buildings within the

update field **20**.

Allocation of Update Cycle Labels

[0066] After establishing zones in which the predictive moving body information is updated for the respective moving bodies **800**, the processing device **100** allocates an update cycle label to the zone established for each moving body **800** based on the moving speed V of the moving body **800**.

[0067] As illustrated in FIG. 6, the processing device **100** allocates ID_1 as an update cycle label to a moving body **800** with a moving speed V of 0.5 kilometers per hour or less. The update cycle of the predictive moving body information of the moving body **800** to which ID_1 is allocated is set to 10 seconds. The processing device **100** allocates ID_2 as an update cycle label to a moving body **800** with a moving speed V greater than 0.5 kilometers per hour and up to 5 kilometers per hour. The update cycle of the predictive moving body information of the moving body **800** to which ID_2 is allocated is set to 1 second. The processing device **100** allocates ID_3 as an update cycle label to a moving body **800** with a moving speed V greater than 5 kilometers per hour and up to 10 kilometers per hour. The update cycle of the predictive moving body information of the moving body **800** to which ID_3 is allocated is set to 0.5 seconds. The processing device **100** allocates ID_4 as an update cycle label to a moving body **800** with a moving speed V greater than 10 kilometers per hour and up to 30 kilometers per hour. The update cycle of the predictive moving body information of the moving body **800** to which ID_4 is allocated is set to 0.2 seconds. The processing device **100** allocates ID_5 as an update cycle label to a moving body **800** with a moving speed V greater than 30 kilometers per hour. The update cycle of the predictive moving body information of the moving body **800** to which ID_5 is allocated is set to 0.1 seconds.

[0068] As shown in FIG. 7, the moving speed V of the first moving body **801** is 50 kilometers per hour. Accordingly, the update cycle label of the first zone AR_1 established for the first moving body **801** is ID_5. The moving speed V of the second moving body **802** is 4 kilometers per hour. Accordingly, the update cycle label of the second zone AR_2 established for the second moving body **802** is ID_2. The moving speed V of the third moving body **803** is 40 kilometers per hour. Accordingly, the update cycle label of the third zone AR_3 established for the third moving body **803** is ID_5. The moving speed V of the fourth moving body **804** is 3 kilometers per hour. Accordingly, the update cycle label of the fourth zone AR_4 established for the fourth moving body **804** is ID_2. The moving speed V of the fifth moving body **805** is 8 kilometers per hour. Accordingly, the update cycle label of the fifth zone AR_5 established for the fifth moving body **805** is ID_3. The moving speed V of the sixth moving body **806** is 4 kilometers per hour. Accordingly, the update cycle label of the sixth zone AR_6 established for the sixth moving body **806** is ID_2. The moving speed V of the seventh moving body **807** is 0.1 kilometers per hour. Accordingly, the update cycle label of the seventh zone AR_7 established for the seventh moving body **807** is ID_1. The moving speed V of the eighth moving body **808** is 20 kilometers per hour. Accordingly, the update cycle label of the eighth zone AR_8 established for the eighth moving body **808** is ID_4. The moving speed V of the ninth moving body **809** is 2 kilometers per hour. Accordingly, the update cycle label of the ninth zone AR_9 established for the ninth moving body **809** is ID_2.

[0069] After allocating the update cycle labels to the zones established for the respective moving bodies **800** based on the moving speeds V of the moving bodies **800**, the processing device **100** advances the process to step S13.

Adjustment of Update Cycles Assigned to the Zones

[0070] In step S13, the processing device **100** adjusts the update cycle of the predictive moving body information determined for each zone in step S11. Specifically, when a part of a zone with a relatively short update cycle and a part of a zone with a relatively long update cycle overlap with each other, the processing device **100** changes the update cycle of the zone with a relatively long update cycle to the update cycle of the zone with a relatively short update cycle.

[0071] As shown in FIG. 5, in the update field **20**, a part of the first zone AR_1 and a part of the

second zone AR_2 overlap with each other. Accordingly, the processing device **100** adjusts the update cycles such that the update cycle of the first zone AR_1 and the update cycle of the second zone AR_2 are the same.

[0072] As shown in FIG. **8**, the moving speed V of the first moving body **801** is 50 kilometers per hour. The moving speed V of the second moving body **802** is 4 kilometers per hour. The update cycle label of the first moving body **801** is ID_5. The update cycle label of the second moving body **802** is ID_2. In the process of step S13, the processing device **100** changes the update cycle label of the second zone AR_2 to ID_5 in order to change the update cycle of the second zone AR_2 to the update cycle of the first zone AR_1. After adjusting the update cycles, the processing device **100** advances the process to step S14.

Update of Predictive Moving Body Information for Each Moving Body **800**

[0073] In step S14, the processing device **100** updates the predictive moving body information for each zone at the update cycle that is set in accordance with the update cycle label allocated to each zone. Thereafter, the processing device **100** transmits the predictive moving body information updated for each zone to the storage device **200**.

[0074] When the update cycle label allocated to the zone in which the second vehicle **602** illustrated in FIG. **3** is present is ID_3, the update cycle of the predictive moving body information is 0.5 seconds. When the update cycle of the predictive moving body information is 0.5 seconds, the processing device **100** updates the predictive moving body information every 0.5 seconds. The processing device **100** calculates the predictive moving body information of the zone in which the second vehicle **602** is present based on the seventh moving body information DAT_7 at the point in time T_0 . The processing device **100** transmits the calculated predictive moving body information to the storage device **200**. The processing device **100** calculates the predictive moving body information of the zone in which the second vehicle **602** is present based on the eighth moving body information DAT_8 at the point in time T_5 . The processing device **100** transmits the calculated predictive moving body information to the storage device **200**.

[0075] During the period from 0.1 seconds to 0.4 seconds, the processing device **100** does not update the predictive moving body information for the zone in which the second vehicle **602** is present. As a result, the update cycle of the predictive moving body information agrees with the acquisition cycle of the moving body information. Consequently, the processing device **100** does not calculate the first calculation data CAL_1, the second calculation data CAL_2, the third calculation data CAL_3, and the fourth calculation data CAL_4.

[0076] After executing the process of step S14, the processing device **100** ends the series of processes.

Calculation of Predictive Moving Body Information of Update Field **20**

[0077] The second processing circuitry **201** of the storage device **200** generates the predictive moving body information of the entire update field **20** based on the predictive moving body information of each zone acquired from the processing device **100**. The second processing circuitry **201** of the storage device **200** stores the predictive moving body information of the entire update field **20** in the second storage circuitry **202**.

Operation of the Present Embodiment

[0078] The longer the update cycle of predictive moving body information is, the lower the processing load on the information processing system **10** becomes. According to the above-described configuration, the processing load on the information processing system **10** is reduced compared to a case in which the entire update field **20** is continuously updated at the same cycle.

Advantages of the Present Embodiment

[0079] (1) The information processing system **10** can continuously update the predictive moving body information with a reduced processing load. [0080] (2) The processing device **100** of the information processing system **10** allocates one zone to each moving body **800**. The range of the zone established for each moving body **800** encompasses the predicted range within which the

moving body **800** is expected to be located. This configuration allows the processing device **100** to change the update cycle of the predictive moving body information for each moving body **800**. The information processing system **10** can reflect real-world moving body information into predictive moving body information with a reduced processing load. [0081] (3) The communication device **300** of the information processing system **10** acquires the moving speed V of each moving body **800** as the moving body information. The processing device **100** of the information processing system **10** extends the update cycles for zones containing moving bodies **800** with relatively low moving speeds V . The lower the moving speed V of a moving body **800** is, the smaller the positional changes of the moving body **800** become. Therefore, when the moving speed V of a moving body **800** is relatively low, extending the update cycle for the zone in which the moving body **800** is present results in a small deviation between the real-world traffic environment and the predictive moving body information. A longer update cycle reduces the processing load on the processing device **100** of the information processing system **10**. This enables the information processing system **10** to calculate the predictive moving body information with a reduced processing load according to the moving speed V of the moving body **800**. [0082] (4) The communication device **300** of the information processing system **10** acquires the moving speed V of each moving body **800** as the moving body information. The processing device **100** of the information processing system **10** narrows zones containing moving bodies **800** with relatively low moving speeds V . When the moving speed V of a moving body **800** is relatively low, the positional changes of the moving body **800** are smaller than those of moving bodies **800** with a relatively high speed. That is, when the moving speed V of a moving body **800** is relatively low, even if the zone assigned to the moving body **800** is narrowed, the moving body **800** will not move beyond the boundaries of that zone. The narrower the zone updated by the processing device **100** of the information processing system **10** is, the lower the processing load on the processing device **100** of the information processing system **10** becomes. This allows the information processing system **10** to reduce the processing load in accordance with the moving speed V of the moving body **800** in the real-world traffic environment. [0083] (5) The processing device **100** of the information processing system **10** acquires information on the traveling direction of each moving body **800** as the moving body information. Based on the moving body information, the processing device **100** calculates a predicted path for each moving body **800**. The processing device **100** establishes the zone of the moving body **800** such that the centroid of the zone is positioned on the side on which the predicted path extends with respect to the center of the moving body **800**. This configuration allows the processing device **100** of the information processing system **10** to update the predictive moving body information in the direction of travel of the moving body **800** without imposing an excessive processing load. [0084] (6) When providing traffic services based on predictive moving body information, it may be necessary to determine the distances between multiple moving bodies **800**. However, if the update cycles of the predictive moving body information differ between moving bodies **800**, accurately determining the distances between the moving bodies **800** is difficult. When a zone with a relatively short update cycle overlaps at least partially with a zone with a relatively long update cycle, the processing device **100** of the information processing system **10** aligns the update cycles by setting both zones to the shorter cycle. As a result, the update cycles of the predictive moving body information for the multiple moving bodies **800** located close to each other agree with each other. This allows the information processing system **10** to accurately reflect the positional relationships between the moving bodies **800** located near each other into the predictive moving body information. [0085] (7) In the real world, there are areas where the vehicles **600** cannot enter, such as locations occupied by buildings. The processing device **100** of the information processing system **10** does not establish zones in locations where the vehicles **600** cannot enter. This configuration reduces the processing load on the processing device **100** of the information processing system **10**. [0086] (8) When the update cycle of the predictive moving body information is different from the acquisition cycle of the moving body information, the processing

device **100** needs to calculate the calculation data CAL based on the moving body information. When the processing device **100** of the information processing system **10** calculates the calculation data CAL, a processing load due to the calculation of the calculation data CAL is applied to the processing device **100**. The processing device **100** can match the update cycle of the predictive moving body information with the acquisition cycle of the moving body information by adjusting the update cycle of the predictive moving body information. In this case, the processing device **100** does not need to calculate calculation data CAL. The processing device **100** does not calculate the calculation data CAL when the calculation data CAL is not necessary. Thus, the processing load on the processing device **100** is reduced compared to a case in which the calculation data CAL is always calculated. [0087] (9) The information processing method executed by the information processing system **10** includes a step (step **S10**) in which the communication device **300** acquires moving body information, which indicates the positions of multiple moving bodies **800** in the real world. The information processing method executed by the information processing system **10** also includes a step (step **S11**) in which the processing device **100** sets the update cycle of predictive moving body information of at least one of multiple zones to be longer than the update cycles of predictive moving body information of other zones. The information processing method executed by the information processing system **10** includes a step (step **S14**) in which the processing device **100** acquires, at a prescribed cycle, predictive moving body information, which is information indicating the position of each of the multiple moving bodies **800** after the point in time at which the moving body information is acquired, based on the moving body information. The longer the update cycle of predictive moving body information is, the lower the processing load on the information processing system **10** becomes. The execution of this information processing method reduces the processing load on the information processing system **10** as compared to a case in which the entire update field **20** is continuously updated at the same cycle. The information processing system **10** can continuously update the predictive moving body information with a reduced processing load. [0088] (10) The first storage circuitry **102** of the processing device **100** of the information processing system **10** stores an information processing program that causes the first processing circuitry **101** of the processing device **100** to execute processes. The information processing program includes causes the first processing circuitry **101** of the processing device **100** to set the update cycle of predictive moving body information in at least one of multiple zones to be longer than the update cycles of predictive moving body information in other zones. The longer the update cycle of predictive moving body information is, the lower the processing load on the information processing system **10** becomes. The above-described information processing program reduces the processing load on the information processing system **10** compared to a case in which the entire update field **20** is continuously updated at the same cycle. The information processing system **10** can continuously update the predictive moving body information with a reduced processing load.

MODIFICATIONS

[0089] The above-described embodiment may be modified as follows. The above-described embodiment and the following modifications can be combined as long as the combined modifications remain technically consistent with each other.

[0090] The processing device **100** of the information processing system **10** may establish a zone in which a moving body **800** is present and a zone in which no moving body **800** is present in the update field **20**. In this case, the processing device **100** sets the update cycle of the predictive moving body information in the zone in which no moving body **800** is present to be longer than the update cycle of the predictive moving body information in the zone in which a moving body **800** is present. The real-world traffic environment in a zone in which no moving body **800** is present is less likely to change compared to a zone in which a moving body **800** is present. Therefore, even if the update cycle for a zone in which no moving body **800** is present is extended, a deviation between the real-world traffic environment and the predictive moving body information is unlikely

to occur. Accordingly, the information processing system **10** can continuously update the predictive moving body information with reduced processing load without causing a large deviation between the real-world traffic environment and the predictive moving body information.

[0091] The processing device **100** of the information processing system **10** may set multiple zones in the update field **20** so that multiple moving bodies **800** are present in one zone. For example, in FIG. 5, one zone in which both the first moving body **801** and the third moving body **803** are present may be established.

[0092] The processing device **100** of the information processing system **10** may define a predicted range within which a specific moving body **800** is expected to be present as a zone in which the specific moving body **800** is present. If another moving body **800** is present and located near the specific moving body **800**, the processing device **100** of the information processing system **10** may expand the zone in which the specific moving body **800** is present beyond the range within which the specific moving body **800** is expected to be present. For example, the processing device **100** may expand the zone in which the specific moving body **800** is present so that it overlaps with a zone in which another nearby moving body **800** is present. Accordingly, the processing device **100** can adjust the update cycle of the zone in which the specific moving body **800** is present and the update cycle of the zone in which another nearby moving body **800** is present on condition that these zones overlap with each other. The processing device **100** may also expand the zone in which a specific moving body **800** is present to a range that does not overlap with a zone in which another nearby moving body **800** is present. In some cases, the update cycle for a zone in which no moving body **800** is present is set to be longer than the update cycle for a zone in which a moving body **800** is present. By expanding the zone in which the moving body **800** is present, the processing device **100** prevents the creation of a zone with a longer update cycle between the zone in which the moving body **800** is present and a zone in which another nearby moving body **800** is present.

[0093] When multiple moving bodies **800** are present in a single update field **20**, the processing device **100** of the information processing system **10** may define, in the update field **20**, zones with different update cycles based on the moving speeds V of moving bodies **800**. For example, the processing device **100** may define a first zone AR_1 and a second zone AR_2 in the update field **20**. The first zone AR_1 of the modification is a zone in which no moving body **800** with a moving speed V higher than or equal to a prescribed first moving speed VL_1 is present. The second zone AR_2 of the modification is a zone in which a moving body **800** with a moving speed V higher than or equal to the prescribed first moving speed VL_1 is present. The processing device **100** sets the update cycle for the first zone AR_1 to be longer than the update cycle of the second zone AR_2. This allows the processing device **100** to change the update cycle of the predictive moving body information in accordance with the moving speed V of each moving body **800** with a processing load less than that in a case in which the update cycle labels are allocated to zones established for the respective moving bodies **800** based on the moving speeds V of the moving bodies **800**. The processing device **100** of the information processing system **10** may establish three or more zones in which the update cycle is set in advance with reference to the moving speeds V of moving bodies **800** in a single update field **20**. For example, the processing device **100** of the information processing system **10** may further establish a third zone AR_3 as a zone in which the update cycle is determined according to the moving speed V of a moving body **800**. The third zone AR_3 of the modification is a zone in which a moving body **800** with a moving speed V higher than or equal to a prescribed second moving speed VL_2 is present. If the third zone AR_3 of the modification is established, the processing device **100** defines the second zone AR_2 of the modification as a zone in which a moving body **800** with a moving speed V that is higher than or equal to the prescribed first moving speed VL_1 and is lower than the prescribed second moving speed VL_2 is present. Further, the processing device **100** sets the update cycle for the second zone AR_2 of the modification to be longer than the update cycle of the third zone AR_3 of the modification.

[0094] If the update field **20** includes a zone having a longer update cycle than other zones, the processing load on the information processing system **10** is reduced by an amount corresponding to the zone having the longer update cycle, as compared to a case in which the update cycle of the entire update field **20** is the same. Therefore, the processing device **100** of the information processing system **10** may set the length of the update cycle of the predictive moving body information regardless of whether a moving body **800** is present.

[0095] The processing device **100** of the information processing system **10** does not necessarily need to change the update cycle for each zone according to the moving speed V of the moving body **800** present in the zone as long as the update cycle of at least one zone is set to be longer than the update cycles of the other zones.

[0096] The processing device **100** of the information processing system **10** may establish multiple zones within the update field **20** to include zones in which no moving body **800** is present.

[0097] When predictive moving body information can be calculated with sufficient accuracy, the processing device **100** of the information processing system **10** may set update cycles regardless of the moving speeds V of the moving bodies **800**.

[0098] When predictive moving body information can be calculated with sufficient accuracy, the processing device **100** of the information processing system **10** may determine the size of each zone regardless of the moving speed V of the moving body **800**.

[0099] When predictive moving body information can be calculated with a sufficient accuracy, the processing device **100** of the information processing system **10** may set the size of each zone regardless of the traveling direction of the moving body **800**.

[0100] In step **S13**, the condition for the processing device **100** of the information processing system **10** to adjust the update cycle is not limited to the case in which two different zones overlap with each other. For example, the processing device **100** may adjust the update cycle on condition that the distance between two different zones is shorter than a prescribed distance.

[0101] Various changes in form and details may be made to the examples above without departing from the spirit and scope of the claims and their equivalents. The examples are for the sake of description only, and not for purposes of limitation. Descriptions of features in each example are to be considered as being applicable to similar features or aspects in other examples. Suitable results may be achieved if sequences are performed in a different order, and/or if components in a described system, architecture, device, or circuit are combined differently, and/or replaced or supplemented by other components or their equivalents. The scope of the disclosure is not defined by the detailed description, but by the claims and their equivalents. All variations within the scope of the claims and their equivalents are included in the disclosure.

Claims

1. An information processing system, comprising processing circuitry, the processing circuitry being configured to: acquire moving body information that indicates a position of each of multiple moving bodies existing in the real world; acquire, based on the moving body information, predictive moving body information at a prescribed cycle, the predictive moving body information indicating a predictive position of each of the multiple moving bodies after a point in time at which the moving body information is acquired; and set the cycle for acquiring the predictive moving body information in at least one zone among multiple zones in which the moving bodies are present to be longer than the cycle for acquiring the predictive moving body information in the other zones.

2. The information processing system according to claim 1, wherein the processing circuitry is configured to: establish a zone in which no moving body is present in addition to the zones in which the moving bodies are present; and set the cycle for the zone in which no moving body is present to be longer than the cycle for the zones in which the moving bodies are present.

3. The information processing system according to claim 1, wherein the moving body information includes a moving speed of each of the moving bodies, the zones include a first zone and a second zone, the first zone is a zone in which no moving body with a moving speed higher than or equal to a prescribed moving speed is present, the second zone is a zone in which one or more moving bodies with a moving speed higher than or equal to the prescribed moving speed are present, and the processing circuitry is configured to set the cycle for the first zone to be longer than the cycle for the second zone.
4. The information processing system according to claim 1, wherein the processing circuitry is configured to allocate one of the zones to each of the moving bodies.
5. The information processing system according to claim 4, wherein the moving body information includes a moving speed of each of the moving bodies, and the processing circuitry is configured to extend the cycle for the zone as the moving speed of the corresponding moving body decreases.
6. The information processing system according to claim 4, wherein the moving body information includes a moving speed of each of the moving bodies, and the processing circuitry is configured to narrow the zone as the moving speed of the corresponding moving body decreases.
7. The information processing system according to claim 4, wherein the moving body information includes information on a traveling direction of each of the moving bodies, and the processing circuitry is configured to calculate a predicted path for each moving body from the moving body information; and define each of the zones for the corresponding moving body such that a centroid of the zone in which the moving body is present is positioned on a side on which the predicted path extends with respect to the center of the moving body.
8. The information processing system according to claim 5, wherein the processing circuitry is configured to, when a zone having a relatively short cycle and a zone having a relatively long cycle at least partially overlap with each other, change the cycle for the zone having the relatively long cycle to the cycle of the zone having the relatively short cycle.
9. The information processing system according to claim 6, wherein the processing circuitry is configured to, when a zone having a relatively short cycle and a zone having a relatively long cycle at least partially overlap with each other, change the cycle for the zone having the relatively long cycle to the cycle of the zone having the relatively short cycle.
10. The information processing system according to claim 7, wherein the processing circuitry is configured to, when a zone having a relatively short cycle and a zone having a relatively long cycle at least partially overlap with each other, change the cycle for the zone having the relatively long cycle to the cycle of the zone having the relatively short cycle.
11. An information processing method, comprising: acquiring moving body information that indicates a position of each of multiple moving bodies existing in the real world; acquiring, based on the moving body information, predictive moving body information at a prescribed cycle, the predictive moving body information indicating a predictive position of each of the multiple moving bodies after a point in time at which the moving body information is acquired; and setting the cycle for acquiring the predictive moving body information in at least one zone among multiple zones in which the moving bodies are present to be longer than the cycle for acquiring the predictive moving body information in the other zones.
12. A non-transitory computer readable storage medium storing an information processing program to be executed by processing circuitry included in an information processing system, wherein the information processing program causes the processing circuitry to set a cycle for acquiring predictive moving body information in at least one zone among multiple zones in which the moving bodies are present to be longer than the cycle for acquiring the predictive moving body information in the other zones, and the predictive moving body information is acquired based on moving body information that indicates a position of each of multiple moving bodies present in the real world, the predictive moving body information indicating a predictive position of each of the multiple moving bodies after a point in time at which the moving body information is acquired.

